**Definition 1.** Let $Var$ be the set of propositional variables. Then $\phi$ is defined inductively as follows:

$$\phi ::= p \mid \phi \vee \phi \mid X\phi \mid \phi U \psi \mid \langle a \rangle \phi$$

**Definition 2.** Let $\phi$ be a formula and $p \in Var$. Then the Closure of $\phi$ is inductively defined as

$$\begin{aligned} Cl(p) &= Cl(\neg p) = \{p, \neg p\} \\ Cl(\neg\phi) &= Cl(\phi) \\ Cl(\phi \vee \psi) &= \{\phi \vee \psi, \neg(\phi \vee \psi)\} \cup Cl(\phi) \cup Cl(\psi) \\ Cl(X\phi) &= \{X\phi, \neg X\phi\} \cup Cl(\phi) \\ Cl(\phi U \psi) &= \{\phi U \psi, \neg(\phi U \psi)\} \cup Cl(\phi) \cup Cl(\psi) \\ Cl(\langle a \rangle \phi) &= \{\langle a \rangle \phi, \neg \langle a \rangle \phi\} \cup Cl(\phi) \end{aligned}$$

**Definition 3.** An atom $A$ is a subset of $Cl(\phi)$ such that

$$\begin{array}{ll} \text{if } \psi \in Cl(\phi), & \text{then } \psi \in A \text{ iff } \neg\psi \notin A, \\ \text{if } \phi_1 \vee \phi_2 \in Cl(\phi), & \text{then } \phi_1 \vee \phi_2 \in A \text{ iff } \phi_1 \in A \text{ or } \phi_2 \in A \\ \text{if } \phi_1 U \phi_2 \in Cl(\phi), & \text{then if } \phi_1 U \phi_2 \in A \text{ then } \psi_1 \in A \text{ or } \psi_2 \in A \\ \text{if } \phi_1 U \phi_2 \in Cl(\phi), & \text{then if } \phi_2 \in A \text{ then } \phi_1 U \phi_2 \in A \end{array}$$

**Definition 4.** We say two atoms $A, B$ are in relation $R_x (AR_xB)$ iff

$$\begin{array}{ll} \text{if } X\psi \in Cl(\phi), & \text{then } X\psi \in A \text{ iff } \psi \in B, \\ \text{if } \psi_1 U \psi_2 \in Cl(\phi), & \text{then if } \psi_1 U \psi_2, \neg\psi_2 \in A, \text{ then } \psi_1 U \psi_2 \in B, \\ & \text{and if } \psi_1 U \psi_2 \in B, \psi_1 \in A, \text{ then } \psi_1 U \psi_2 \in A. \end{array}$$

**Lemma 5.** *For any atom $A$ there exists an atom $B$ s.t. $AR_xB$.*

*Proof.* Assume $A$ is an atom. Then we define $B'$ as follows:

$$B' = \{\phi \mid X\phi \in A\} \cup \{\neg\phi \mid \neg X\phi \in A\} \cup \{\psi_1 U \psi_2 \mid \psi_1 U \psi_2, \neg\psi_2 \in A\} \cup \{\neg(\psi_1 U \psi_2) \mid \neg(\psi_1 U \psi_2), \psi_1 \in A\}$$

Now we show that $B$ is consistent. Assume it is not. Then there exists $\alpha_1, ..., \alpha_n$ s.t. $B \vdash \alpha_1 \wedge \alpha_2 ... \wedge \alpha_n \to \bot$. Let's define $\hat{B} = \bigwedge B$. It follows:

$$\begin{array}{ll} \vdash \hat{B} \wedge \alpha_1 \wedge ... \wedge \alpha_n \to \bot & \\ \vdash X(\hat{B} \wedge \alpha_1 \wedge ... \wedge \alpha_n \to \bot) & \text{By X-necessitation} \\ \vdash X(\hat{B} \wedge \alpha_1 \wedge ... \wedge \alpha_n) \to X\bot & \text{F2} \\ \vdash X(\hat{B}) \wedge X\alpha_1 \wedge ... \wedge X\alpha_n \to X\bot & \end{array}$$

Assume, $a_i = \phi$. Then by definition we have $A \vdash Xa_i$. If $a_i = \neg\phi$, then we can use F1 to obtain $A \vdash X\neg\phi$. If $a_i = \psi_1 U \psi_2$ then by definition, $\psi_1 U \psi_2 \in A$ and $\neg\psi_2 \in A$. Consider the following proof steps:

| | |
|---|---|
| $A \vdash \neg\psi_2$ | |
| $A \vdash \psi_1 U \psi_2$ | |
| $A \vdash \psi_1 U \psi_2 \wedge \neg\psi_2$ | |
| $A \vdash \psi_1 U \psi_2 \wedge \neg\psi_2 \rightarrow (\psi_2 \vee (\psi_1 \wedge X(\psi_1 U \psi_2))) \wedge \neg\psi_2$ | Unfolding U |
| $A \vdash \psi_1 U \psi_2 \wedge \neg\psi_2 \rightarrow (\psi_1 \wedge X(\psi_1 U \psi_2))$ | distributivity and other rules |
| $A \vdash \psi_1 \wedge X(\psi_1 U \psi_2)$ | |
| $A \vdash X(\psi_1 U \psi_2)$. | |

Hence, $A \vdash Xa_i$. If $a_i = \neg(\psi_1 U \psi_2)$, then by definition, $\neg(\psi_1 U \psi_2), \psi_1 \in A$. Our goal is to derive $A \vdash \neg(\psi_1 U \psi_2) \wedge \psi_1 \rightarrow X\neg(\psi_1 U \psi_2)$.

| | |
|---|---|
| $A \vdash \neg(\psi_1 U \psi_2) \wedge \psi_1$ | |
| $A \vdash \neg(\psi_2 \vee (\psi_1 \wedge X(\psi_1 U \psi_2))) \wedge \psi_1$ | Unfolding U |
| $A \vdash \neg\psi_2 \wedge (\neg\psi_1 \vee \neg X(\psi_1 U \psi_2)) \wedge \psi_1$ | |
| $A \vdash \neg\psi_1 \vee \neg X(\psi_1 U \psi_2)$ | |
| $\neg X(\psi_1 U \psi_2)$ | $\psi_1$ is derivable |
| $A \vdash X\neg(\psi_1 U \psi_2)$ | F1 |

By similar argument, we can conclude $A \vdash X\hat{B}$. So it follows

$$A \vdash X\bot$$

Because $X\bot \rightarrow \bot$ is a tautology, we conclude $A \vdash \bot$ which is a contradiction. Now, by the Extension Lemma, we can extend $B'$ to an atom $B$. It is easy to check, that $AR_xB$.

□

**Lemma 6.** *Let $\Gamma = \{\phi\}'$ be a consistent set of formulas. Then $\Gamma'$ can be extended to an atom.*

*Proof.* Let $n = |CL(\phi)|$ and consider the enumeration $\psi_1, \psi_2, ..., \psi_n$ of $CL(\phi)$. We define $\Gamma_0 = \{\phi\}$ and then inductively

$$\Gamma_{n+1} := \begin{cases} \Gamma_n \cup \psi_{n+1}, & \text{if } \Gamma_n \cup \psi_{n+1} \text{ is consistent,} \\ \Gamma_n \cup \neg\psi_{n+1}, & \text{else .} \end{cases}$$

Then $\Gamma^+ = \bigcup_n \Gamma_n$ is our atom. Idea: Show for each $\Gamma_n$ is consistent for each $n$.

□

**Lemma 7.** *Assume $\psi_1 U \psi_2 \in A_i$. Then there is an $n \in \mathbb{N}$ with $A_i R_x A_{i+1} R_x ... R_x A_n$ s.t. $\psi_2 \in A_n$ and for all $i \leq j < n : \psi_1 \in A_j$.*

**Definition 8.** Let $G_i$ be some global atom. We say an atom B is $G_i$-allowed if $G_i \vdash \neg\langle a\rangle\psi$ and $\psi \in Cl(\phi)$, then $\neg\psi \in B$

**Lemma 9.** *Assume a global path $G_0 G_1 ...$, where $G_0$ is the initial state. Then for $\langle a\rangle\phi \in G_0$ we can find some $G_0$-admissible atom $A$ with $\langle a\rangle\phi \in A$.*

Here we construct for each $\langle a\rangle\phi$ an intial state in $T_a$.

But the idea doesnt work yet. The constructed $T_a$ could contain not allowed atoms anymore.

**Definition 10.** Assume a global path $G_0 G_1 ...$ and some $A_0 A_1 ...$ that is allowed w.r.t. to their position. Then we define $(n, j, A_j) R_a (n', j', A_k)$ iff $A_j R_x A'_j$, $j' = j + 1$ and $n = n'$ where n and n' will be the 'ID' of the path and j,j' are the atom positions in the global path s.t. $A_j$ or $A'_j$ is allowed respectively.

**Lemma 11.** *Assume we have a global path $G_0 A_1 G_2 ....$ If $\langle a\rangle\phi \in A_i$ then there exists $A_0, A_1, ..., A_i$ with $(0, A_0) R_a (1, A_1) ... R_a (i, A_i)$ s.t. $\phi \in A_i$.*

**Lemma 12.** *Assume some path $A_0 A_1 ... A_i$ in $T_a$ with $A_0 R_a A_1 ... R_a A_i$ s.t $\phi \in A_i$. Then there exists an infinite fulfilling extension s.t. it is allowed w.r.t to the global path.*

**Lemma 13.** *Assume the atom $A_0$. Then there exists an infinite fulfilling path that extends $A_0$, i.e. there is some path $A_0 A_1 ...$ with $A_i R_x A_{i+1}$ for all $i \in \mathbb{N}$ s.t. for any $\psi_1 U \psi_2 \in A_i$ : there exists an $j \geq i$ s.t. $\psi_2 \in A_j$ and for all $i \leq k < j$ $\psi_1 \in A_k$*

**Lemma 14.** *Assume some path $A_0 ... A_i$ in $T_a$ that is allowed. Then this path can be extended to an infinite allowing path $A_{i+1} A_{i+2} - - -$s.t. $\forall n \geq i \forall \psi_1 U \psi_2 \in A_n \exists j$ s.t. $\psi_2 \in A_j$ and $\forall i \leq k < j$: $\psi_1 \in A_k$.*

**Lemma 15.** *Assume a global path $G_0 G_1 ....$ Then for any $\psi \in Cl(\phi)$:*

$$\psi \in G_i \text{ iff } ((G_i G_{i+1} ...), (G_1 ... G_i)) \vDash \psi$$

*For every agent a, every $A_0 A_1 ...$ path in $T_a$ and for any $\psi \in Cl(\phi)$:*

$$\psi \in A_i \text{ iff } ((A_i A_{i+1} ...), (A_1 ... A_i)) \vDash \psi$$

*Proof.* We show both on double induction on the length of $\phi$. We first prove the global case. Assume $\phi = p$. We define the truth on each atom to be $L(G_i) = \{p \mid p \in G_i\}$. Then it follows:

$$p \in G_i \iff p \in L(G_i) \iff (G_i...G_\infty, G_0...G_i) \vDash p$$

. If $\phi = \neg\psi$, then we have:

$$\neg\psi \in G_i \iff \psi \notin G_i \overset{\text{IH}}{\iff} (G_i...G_\infty, G_0...G_i) \nvDash \psi \iff (G_i...G_\infty, G_0...G_i) \vDash \neg\psi$$

. $\phi = \psi_1 \vee \psi_2$ is done analogously. Assume $\phi = X\psi \in G_i$. Then $X\phi \in G_i \iff \psi \in G_{i+1} \overset{\text{IH}}{\iff} (G_{i+1}...G_\infty, G_0...G_{i+1}) \vDash \psi \iff (G_i...G_\infty, G_0...G_i) \vDash X\psi$.

Next we deal with the Until case, so assume $\phi = \psi_1 U \psi_2$. We prove two directions. If $\psi_1 U \psi_2 \in G_i$, then by construction of the global path there exists some $j \geq i$ s.t. $\psi_2 \in G_j$ and for all $i \leq k < j$: $\psi_1 \in G_k$. By IH, it follows $(G_j...G_\infty, G_0...G_j) \vDash \psi_2$ and $(G_k...G_\infty, G_0...G_k) \vDash \psi_1$. Hence, $(G_i...G_\infty, G_0...G_i) \vDash \psi_1 U \psi_2$. Now assume $(G_i...G_\infty, G_0...G_i) \vDash \psi_1 U \psi_2$. By the semantics that means there is some $j \geq i$ s.t. $(G_j...G_\infty, G_0...G_j) \vDash \psi_2$ and for all $i \leq k < j$: $(G_k...G_\infty, G_0...G_k) \vDash \psi_1$. Now IH gives us $\psi_2 \in G_j$ and hence $\psi_1 U \psi_2 \in G_j$. Because $G_{j-1} R_x G_j$, and by IH $\psi_1 \in G_{j-1}$, we conclude $\psi_1 U \psi_2 \in G_{j-1}$. Applying this argument repeatedly, we get $\psi_1 U \psi_2 \in G_i$.

□